C++ Midterm Practice: 100 Questions

Savitch, *Problem Solving with C++*, Chapters 1–5 · Multiple choice · Answer key with explanations at the end

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Chapter 1 Computers and C++ Programming

1. Which of these is NOT one of the five main components of a computer?

A. Processor (CPU)
B. Main memory
C. Input devices
D. Compiler

2. A binary digit (bit) is:

A. The address of a memory location
B. A digit that can only be 0 or 1
C. A group of eight binary values
D. A single character such as 'A'

3. How many bits are in one byte?

A. 4
B. 32
C. 8
D. 16

4. A data item that is too large for one byte is stored in several consecutive bytes. Its address refers to:

A. Every byte equally
B. The middle byte
C. The last byte
D. The first byte

5. The same bit pattern `01000001` could mean 'A', 65, or an instruction. How does the computer decide?

A. By the file extension
B. The operating system labels every byte
C. By the current instruction that uses it
D. The pattern always means 'A'

6. Which memory holds instructions and data between sessions (after power is off)?

A. Main memory
B. CPU registers
C. Secondary memory
D. Cache

7. “Random access” memory means:

A. Data is stored in random order
B. Data must be searched sequentially
C. The computer can directly access any memory location
D. Contents change randomly

8. A compiler translates:

A. A high-level language program into machine language
B. Machine language into a high-level language
C. Assembly into a high-level language
D. Object code into source code

9. The original program written in a high-level language is called the:

A. Executable
B. Source code
C. Machine code
D. Object code

10. What does a linker do?

A. Finds syntax errors
B. Loads the operating system
C. Combines your object code with object code for precompiled routines (such as I/O) into a runnable program
D. Translates C++ into assembly

11. Who is considered the first programmer?

A. Grace Hopper
B. Charles Babbage
C. Ada Augusta, Countess of Lovelace
D. Bjarne Stroustrup

12. An algorithm is:

A. Any C++ source file
B. A sequence of precise instructions that leads to a solution
C. A program written in machine language
D. A type of compiler

13. The two main phases of program design are:

A. Input and output
B. Problem-solving phase and implementation phase
C. Testing and maintenance
D. Compiling and linking

14. In OOP, a single name having multiple meanings depending on its context is called:

A. Inheritance
B. Encapsulation
C. Abstraction
D. Polymorphism

15. A program compiles and runs without crashing but prints the wrong answer. This is a:

A. Linker error
B. Logic error
C. Run-time error
D. Syntax error

Chapter 2 C++ Basics

16. Which is a valid C++ identifier?

A. double
B. my-total
C. 2ndValue
D. _total2

17. Keywords (reserved words) in C++:

A. Can be used as variable names if capitalized the same way
B. Cannot be used as identifiers
C. Must be declared before use
D. Are only used in comments

18. What does `numberOfBars = numberOfBars + 3;` do?

A. It is a compile error because it can't be true
B. Compares the two sides
C. Sets numberOfBars to its previous value plus 3
D. Sets numberOfBars to 3

19. After `double mpg(26.3), area(0.0), volume;` which statement is true?

A. mpg is 26.3, area is 0.0, volume is uninitialized
B. volume is 0.0 by default
C. It is a syntax error
D. All three are initialized

20. What is printed?

```
cout << 17 / 5 << " " << 17 % 5;
```

A. 3 2
B. 3.4 0
C. 3 3
D. 3.4 2

21. What value does `d` hold?

```
double d = 11 / 2;
```

A. 5.0
B. Compile error
C. 6.0
D. 5.5

22. What is the value of `7.0 / 2` ?

A. 3.5
B. 4 D.
C. 3
3.0

23. What is the value of `10 / 4 * 4` ?

A. 1
B. 8
C. 10
D. 2.5

24. What does `i` hold?

```
int i = 7.9;
```

A. Compile error
B. 7
C. 7.9
D. 8

25. What is the value of `n` ?

```
int n = 4;
n *= 2 + 3;
```

A. 14
B. 20
C. 11
D. 9

26. Which escape sequence produces a tab?

A. \"
B. \n
C. \\
D. \t

27. Which three statements make `cout` always show exactly two decimal places for doubles?

A. cout.precision(2); only
B. cout << fixed2;
C. cout.setf(ios::fixed); cout.setf(ios::showpoint); cout.precision(2);
D. cout.setf(ios::decimal, 2);

28. `#include <iostream>` is needed to:

A. Use the string class
B. Enable math functions
C. Declare main
D. Make the definitions of cin and cout available

29. With `cin >> v1 >> v2 >> v3;` the user's three values should be separated by:

A. Semicolons
B. Tabs only
C. Spaces (whitespace)
D. Commas

30. Which is a constant of type `char` ?

A. 'A'
B. A
C. "A"
D. char(A)

31. To use the `string` class you must include:

A. <string>
B. <cstring> only
C. <iostream>
D. <cmath>

32. `5.89e-6` equals:

A. 5890000
B. 0.00000589
C. 0.0000589
D. -5.89

33. What is printed?

```
int x = 1;
while (x < 20)
    x = x * 2;
cout << x;
```

A. 16
B. 32
C. Infinite loop
D. 20

34. What is printed?

```
int n = 10;
do {
    cout << n << " ";
    n++;
} while (n < 5);
```

A. Nothing
B. 10
C. Infinite loop
D. 10 11 12 13 14

35. What is wrong with `if (x = 5) cout << "five";` ?

A. It is a syntax error

B. It assigns 5 to x, so the condition is always true

C. Nothing

D. It compares x to 5 twice

36. Given `const double PI = 3.14159;` , what happens with `PI = 3.14;` later?

A. Run-time error

B. PI becomes 3.14

C. Compile error

D. Warning only; PI stays 3.14159

37. Which type holds only the values `true` and `false` ?

A. enum

B. int

C. bool

D. char

Chapter 3 More Flow of Control

38. If `y` is 8, what is `!( (y < 3) || (y > 7) )` ?

A. true

B. false

C. Compile error

D. 8

39. Without parentheses, the default precedence order (highest first) is:

A. ||, &&, relational, !

B. relational, !, ||, &&

C. !, relational, &&, ||

D. &&, ||, !, relational

40. With `kids = 0` , what happens in `if ((kids != 0) && (pieces / kids >= 2))` ?

A. Short-circuit evaluation skips the division; condition is false

B. Division-by-zero run-time error

C. Condition is true

D. Compile error

41. With `time = 36` and `limit = 60` , what is `(!time > limit)` ?

A. true, because 36 is not greater than 60

B. false, because it is evaluated as (!time) > limit, i.e. 0 > 60

C. Undefined

D. A compile error

42. When an integer is used as a Boolean value:

A. Any nonzero value is true; 0 is false

B. It causes an error

C. Negative numbers are false

D. Only 1 is true

43. Given `enum Direction {NORTH, SOUTH, EAST, WEST};` what is the value of EAST?

A. 'E'

B. 1

C. 2

D. 3

44. Given `enum MyEnum {ONE = 17, TWO, THREE, FOUR = -3, FIVE};` what is FIVE?

A. 20

B. -2

C. 4

D. -4

45. What is printed when `fuel` is 0.5?

```
if (fuel < 0.75)
    if (fuel < 0.25)
        cout << "Fuel very low.";
    else
        cout << "Fuel over 3/4.";
```

A. Both messages
B. Nothing
C. Fuel over 3/4.
D. Fuel very low.

46. What is printed?

```
int n = 2;
switch (n) {
    case 1: cout << "A";
    case 2: cout << "B";
    case 3: cout << "C"; break;
    default: cout << "D";
}
```

A. BC
B. B
C. BCD
D. ABC

47. The controlling expression of a `switch` may NOT be of type:

A. int
B. char
C. double
D. bool

48. In a switch, the `default` case runs when:

A. It always runs last
B. A break is missing
C. No case label matches
D. The first case matches

49. What is printed?

```
for (int i = 0; i < 10; i += 3)
    cout << i << " ";
```

A. 0 3 6
B. 0 3 6 9
C. 3 6 9
D. 0 3 6 9 12

50. How many times does the body of `for (int i = 1; i <= 5; i++)` execute?

A. 5
B. 6
C. 4
D. Infinite

51. After this code, what are m and n?

```
int n = 5;
int m = n++ + 2;
```

A. m = 8, n = 5
B. m = 7, n = 6
C. m = 8, n = 6
D. m = 7, n = 5

52. What is printed?

```
for (int i = 0; i < 5; i++);
cout << "hi";
```

A. hi printed 5 times
B. hi printed once
C. Nothing
D. Compile error

53. What is printed?

```
int t = 0;
for (int i = 1; i <= 10; i++)
    { if (i % 2 == 0) continue;
    if (i > 7) break;
    t += i; }
cout << t;
```

A. 25
B. 55
C. 9
D. 16

54. The `break` statement inside a loop:

A. Ends the loop immediately
B. Skips to the next iteration
C. Ends the program
D. Restarts the loop

55. Which loop is the natural choice when the body must execute at least once?

A. while
B. for
C. switch
D. do-while

56. A loop that stops when it reads a special value that cannot be valid data uses a:

A. Counter
B. Sentinel value
C. Flag
D. Break-point

57. How many stars are printed?

```
for (int i = 0; i < 3; i++)
    for (int j = 0; j < 4; j++)
        cout << "*";
```

A. 12
B. 9
C. 16
D. 7

58. What is printed?

```
int sum = 0;
for (int i = 1; i <= 4; i++)
    sum += i;
cout << sum;
```

A. 15
B. 6
C. 10
D. 4

59. What happens?

```
int x = 10;
while (x != 0)
    x -= 3;
```

A. Compile error
B. Loop runs 3 times
C. Infinite loop
D. Loop runs 4 times

Chapter 4 Procedural Abstraction & Functions That Return a Value

60. Top-down design is also called:

A. Bottom-up design
B. Overloading
C. Stepwise refinement
D. Procedural abstraction

61. In `theRoot = sqrt(9.0);` the value 9.0 is the function's:

A. Return value
B. Argument
C. Parameter type
D. Prototype

62. Which library must be included to use `sqrt` and `pow`?

A. <cstdlib>
B. <iostream>
C. <cmath>
D. <ctime>

63. Which function returns the absolute value of a `double`?

A. dabs
B. abs
C. fabs
D. labs

64. What is `floor(5.8) + ceil(5.2)`?

A. 10
B. 11
C. 12
D. 5.8

65. What is `7 / abs(-2)`?

A. 3.5
B. -3
C. 3
D. 4

66. What is `pow(2, 3)`?

A. 9
B. 8
C. 6
D. 5

67. Which expression simulates rolling a six-sided die?

A. rand(6)
B. rand() % 6
C. (rand() % 6) + 1
D. rand() % 7

68. `srand(time(0));` should be called:

A. Before every call to rand()
B. After every loop
C. Never; rand() seeds itself
D. Only once, to seed the random number generator

69. With `int totalCandy = 9, numberOfPeople = 4;` which gives 2.25?

A. static_cast<double>(totalCandy / numberOfPeople)
B. static_cast<double>(totalCandy) / numberOfPeople
C. double(totalCandy / numberOfPeople)
D. totalCandy / numberOfPeople

70. A function declaration (prototype):

A. Must appear after main
B. Contains the function body
C. Ends with a semicolon and must appear before the function is called
D. Is only needed for void functions

71. Which statement is true?

A. double totalCost(int, double); is a valid declaration
B. Function headers may omit parameter names
C. A function may be declared only once per program and never defined
D. Declarations must list parameter names

72. What is printed?

```
   int f(int n)
   { n = n * 2;
     return n + 1; }
// in main:
int x = 3;
int y = f(x);
cout << x << " " << y;
```

A. 6 7 C. 6 13
B. 3 7
D. 3 6

73. With call-by-value parameters:

A. The argument variable changes when the parameter changes
B. The & symbol is required
C. Arguments must be variables
D. The formal parameter receives a copy of the argument's value

74. Given `char grade(int received, int minScore);` the call `grade(minScore, received)`:

A. Swaps them automatically
B. Is caught by the compiler
C. Causes a run-time crash
D. Compiles but gives a wrong result because arguments are in the wrong logical order

75. The `return` statement in a value-returning function:

A. Only works in main
B. Prints the value
C. Ends the function call and sends back a value
D. Is optional

76. A function that returns `bool` can be used:

A. Only inside loops
B. Only with call-by-reference
C. Only in cout statements
D. Wherever a Boolean expression is expected, such as an if condition

77. Procedural abstraction means a function should be used like a:

A. Global variable
B. Black box: you need to know what it does, not how
C. Loop
D. Header file

78. A variable declared inside a function body is:

A. A named constant
B. Local to that function
C. Shared with main
D. Global

79. A named constant declared outside all functions (e.g. `const double PI = 3.14159;` above main) is:

A. Illegal
B. Local to main
C. A global named constant, usable by all functions after it
D. Visible only to the next function

80. Two overloaded functions must differ in:

A. Parameter names only
B. The comments above them
C. Number and/or types of parameters
D. Return type only

Chapter 5 Functions for All Subtasks

81. A `void` function:

A. Cannot have parameters
B. Cannot contain return
C. Returns an int by default
D. Returns no value

82. Given `void showResults(double f, double c);` which call is correct?

A. cout << showResults(32.5, 0.3);
B. double r = showResults(32.5, 0.3);
C. showResults(32.5, 0.3);
D. return showResults(32.5, 0.3);

83. Why might a void function contain `return;` ?

A. To return the last variable
B. It is required at the end
C. To return 0
D. To end the function early, e.g. to avoid division by zero

84. What is the bug in `celsius = (5 / 9) * (fahrenheit - 32);` ?

A. fahrenheit must be int
B. Missing parentheses
C. No bug
D. 5 / 9 is integer division and equals 0, so the result is always 0

85. The `&` in `void getInput(double& f);` means:

A. f is a call-by-reference parameter
B. f is global
C. f is a pointer
D. f is a constant

86. The argument for a call-by-reference parameter must be:

A. A variable
B. A number
C. A constant
D. An expression

87. What is printed?

```
void swapValues(int& a, int& b)
    { int t = a; a = b; b = t;
}
// in main:
int x = 4, y = 9;
swapValues(x, y);
cout << x << " " << y;
```

A. 9 4
B. 4 4
C. 9 9
D. 4 9

88. The & was left off both parameters. What is printed?

```
void swapValues(int a, int b)
    { int t = a; a = b; b = t;
}
// in main:
int x = 4, y = 9;
swapValues(x, y);
cout << x << " " << y;
```

A. 9 4
B. Compile error
C. 4 9
D. 9 9

89. In `void goodStuff(int& p1, int p2, double& p3);` which parameters can change the caller's variables?

A. p1 only
B. p1 and p3
C. p2 only
D. All three

90. What is printed?

```
void f(int& a, int b)
    { a = a + b;
    b = b * 2; }
// in main:
int x = 2, y = 5;
f(x, y);
cout << x << " " << y;
```

A. 7 10
B. 7 5
C. 2 10
D. 2 5

91. What is printed?

```
void g(int& a, int& b)
    { a = a + 1;
    b = b + 1; }
// in main:
int n = 1;
g(n, n);
cout << n;
```

A. 1
B. 2
C. 3
D. 4

92. How do you decide between call-by-value and call-by-reference?

A. Use reference if the function must change the argument variable; otherwise value
B. Always use reference for speed
C. Use value for doubles, reference for ints
D. Use reference only in void functions

93. A precondition:

A. Is checked by the compiler
B. Is the function's return type
C. Describes the effect after the call
D. States what is assumed true when the function is called

94. A postcondition:

A. Must be written after main
B. Is optional C++ syntax
C. Describes what will be true after the call, including the return value and changes to reference parameters
D. Lists the function's local variables

95. Which is true about functions calling functions?

A. A function body can call another function, whose declaration must appear before the call
B. Functions can't call void functions
C. Only main can call functions
D. A function can be defined inside another function

96. A driver program is:

- **A.** The operating system
- **B.** A program written only to test a single function
- **C.** A simplified function that returns fixed values
- **D.** The main program of the final product

97. A stub is:

- **A.** A function with no parameters
- **B.** A test program for one function
- **C.** A broken function
- **D.** A simplified version of a not-yet-tested function, used while testing other functions

98. The fundamental rule for testing functions says every function should be tested in a program in which:

- **A.** There are no loops
- **B.** main is empty
- **C.** Every other function has already been fully tested and debugged
- **D.** All functions are stubs

99. What happens when the boolean in `assert(denominator != 0);` is false?

- **A.** It prints a warning and continues
- **B.** denominator is set to 1
- **C.** The compiler rejects the program
- **D.** The program aborts

100. Which is NOT a recommended debugging technique?

- **A.** Randomly changing code until it works
- **B.** Using cout statements to trace variable values
- **C.** Keeping an open mind about where the bug is
- **D.** Using a debugger to step line by line

Answer Key

Correct letter and a short explanation for each question.

Chapter 1

1. D The five components are input devices, output devices, processor, main memory, and secondary memory. A compiler is software.

2. B A bit holds only 0 or 1. Eight bits make a byte.

3. C Each memory location (byte) holds eight bits.

4. D The address of a multi-byte item is the address of its first byte.

5. C Interpretation depends on the instruction currently operating on that memory.

6. C Secondary memory (disk, etc.) is the permanent record; main memory holds the running program.

7. C RAM allows direct access to any location. Sequential access requires passing other items first. **8. A** The compiler turns source code (high-level) into object code (machine language).

9. B Source code is the original; object code is the translated version. **10. C** The linker joins your object code with already-compiled library code.

11. C Ada Lovelace worked with Babbage, who designed the first programmable computer.

12. B A program is an algorithm expressed in a language the computer understands.

13. B Problem solving produces the algorithm; implementation translates it into a programming language. **14. D** Encapsulation is information hiding; inheritance is reuse; polymorphism is one name, many meanings. **15. B** Logic errors are the hardest: the compiler and computer can't detect them.

Chapter 2

16. D Must start with a letter or underscore, contain only letters, digits, underscores, and not be a keyword.

17. B Keywords like int, double, if have fixed meanings and cannot be identifiers.

18. C = is assignment, not equality.

19. A Parentheses form initializes; volume is only declared and holds a garbage value.

20. A Integer division discards the fraction (3); % gives the remainder (2).

21. A 11 / 2 is integer division = 5, then stored as 5.0.

22. A If either operand is double, floating-point division is used.

23. B / and * have equal precedence, left to right: (10/4)*4 = 2*4 = 8.

24. B Assigning a double to an int truncates the fractional part (no rounding).

25. B n *= expr means n = n * (expr) = 4 * 5 = 20.

26. D \n newline, \t tab, \\ backslash, \" double quote.

27. C fixed selects fixed-point notation, showpoint forces the decimal point, precision(2) sets two digits.

28. D iostream holds cin/cout; using namespace std makes the names available.

29. C Values are whitespace-separated and not read until Enter is pressed.

30. A char constants use single quotes; double quotes make a string.

31. A #include along with using namespace std.

32. B e-6 moves the decimal point six places left.

33. B x: 1, 2, 4, 8, 16, 32. The loop stops when x = 32.

34. B A do-while body always runs at least once; then n < 5 is false.

35. B = assigns; == compares. A nonzero value counts as true.

36. C A const named constant cannot be changed after initialization.

37. C bool is the Boolean type.

Chapter 3

38. B !(false || true) = !(true) = false.

39. C ! first, then < > etc., then &&, then ||.

40. A The left side is false, so && doesn't evaluate the right side.

41. B ! has higher precedence than >. Write !(time > limit) if that's what you mean.

42. A C++ treats nonzero as true and zero as false.

43. C Values start at 0 by default: NORTH 0, SOUTH 1, EAST 2, WEST 3.

44. B Each unspecified constant is 1 more than the previous: FOUR is -3, so FIVE is -2.

45. C The else pairs with the nearest previous if (fuel < 0.25). Use braces to fix it.

46. A Without break, execution falls through from case 2 into case 3, then break stops it.

47. C It must be an integral type such as int, char, bool, or an enum. Not double or string.

48. C default handles all values not listed in a case.

49. B i takes 0, 3, 6, 9; then 12 fails i < 10.

50. A i = 1, 2, 3, 4, 5.

51. B n++ returns the old value (5) then increments. m = 5 + 2 = 7, n = 6.

52. B The stray semicolon is an empty loop body. cout runs once after the loop.

53. D Even i are skipped. Odd i: 1+3+5+7 = 16, then i = 9 triggers break.

54. A break exits the loop; continue skips the rest of the current iteration.

55. D do-while tests its condition after the body.

56. B E.g. reading scores until a negative number is entered.

57. A The inner loop runs 4 times for each of 3 outer iterations: 3 × 4 = 12.

58. C 1 + 2 + 3 + 4 = 10.

59. C x goes 10, 7, 4, 1, -2, … and never equals 0. Prefer x > 0 in such tests.

Chapter 4

60. C Break the task into subtasks, then refine each until trivial.

61. B sqrt(9.0) is a function call with argument 9.0; it returns 3.0.

62. C cmath holds sqrt, pow, fabs, ceil, floor.

63. C abs works on int (cstdlib); fabs on double (cmath).

64. B floor(5.8) = 5, ceil(5.2) = 6, total 11.

65. C abs(-2) is int 2, and 7 / 2 is integer division = 3.

66. B 2 raised to the 3rd power is 8.

67. C rand() % 6 gives 0–5; adding 1 gives 1–6.

68. D Seed once so each run produces a different pseudo-random sequence.

69. B Cast before dividing. Casting the result of 9/4 gives 2.0 because integer division already happened.

70. C The prototype tells the compiler the return type, name, and parameter types.

71. A Declarations may list only types; the definition's header must name the parameters.

72. B Call-by-value: n is a copy, so x stays 3. y = 3*2 + 1 = 7.

73. D Changes to the formal parameter don't affect the argument.

74. D The compiler checks types and count, not logical order.

75. C Each branch may have its own return.

76. D e.g. if (appropriate(rate)) … reads more clearly than the full expression.

77. B The declaration and its comment should be all a user needs.

78. B Local variables exist only within the function; same names elsewhere are different variables.

79. C Global named constants are common and safe; global variables are discouraged.

80. C Functions that differ only in return type cannot be overloaded.

Chapter 5

81. D void replaces the return type; return; with no expression may end it early.

82. C A void function call is a statement by itself, not part of an expression.

83. D If omitted, an implicit return happens at the end of the body.

84. D Write 5.0 / 9 to get floating-point division.

85. A The & goes in both the declaration and the definition.

86. A The function receives the memory location of the argument, so it must be a variable.

87. A Reference parameters change the arguments themselves.

88. C Forgetting & makes them call-by-value: inadvertent local variables. x and y don't change.

89. B Only reference parameters (with &) affect the arguments.

90. B a is a reference to x (becomes 7); b is a copy of y, so y stays 5.

91. C Both a and b refer to the same variable n, so it is incremented twice.

92. A That is the rule in Display 5.6.

93. D The function should not be called unless the precondition holds.

94. C Write pre/postconditions before designing how the function works.

95. A Functions cannot be nested, but they can call each other.

96. B Drivers call the function with test data and display the results.

97. D Stubs usually just return test values and are simple enough to trust.

98. C This way any bug found belongs to the function under test.

99. D #include . assert is used to check pre- and postconditions.

100. A Don't change code without understanding what you're doing.